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Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE – 411037.
 (An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Examination: ESE

Year: TY

Branch: IT

Subject: Design and Analysis of Algorithms **Subject Code: IT3215**

Max. Marks: 60

Total Pages of Question Paper: 02

Day & Date:

Time:

Instructions to Candidate

1. All questions are compulsory.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right indicate full marks.

Q. N.	C O N o	BT* No		Max marks
			Question 1. a is compulsory and solve any one out of 1.b and 1.c	
Q. 1. a.	1	3,4	Apply Master theorem to solve the following recurrences: i) $T(n) = 4T(n/2) + n$. ii) $T(n) = 4T(n/2) + n^2$	6
b.	1	1,3,	Define Theta (θ) notation and prove that $4n^2+3n$ belongs to $\theta(n^2)$	4
c.	1	1,4,5	Define Best Case, Average Case, and Worst Case time complexity. Analyze the following sorting algorithm in terms of Best Case, Average Case, and Worst Case time complexity. Identify which sorting algorithm it implements. Justify your answer. <pre>void sort(int a[], int n) { int i, j, temp; for (i = 0; i < n - 1; i++) { for (j = 0; j < n - i - 1; j++) { if (a[j] > a[j + 1]) { temp = a[j]; a[j] = a[j + 1]; a[j + 1] = temp; } } } }</pre>	4
			Question 2. a is compulsory and solve any one out of 2.b and 2.c	
Q. 2. a.	2	1,2,4,5	Explain a control abstraction for divide and conquer strategy. Write recurrence relations for Quick sort(Best/Average Case and Worst case) and solve recurrence relation using substitution method and derive it's time complexity.	6
b.	2	2,3,4	You have Quick Sort and Merge Sort available. For the given data sets, determine which sorting algorithm you would choose to sort the data in ascending order . Justify your answer using Big-O notation and explain why the selected algorithm is better for each case. Data Sets: a) Set-1: {0,1,2,3,4,5,6,7,8,9,10} b) Set-2: {10,9,8,7,6,5,4,3,2,1,0} c) Set-3: {0,2,4,3,5,7,8,6,9,10,1}	4
c.	2	1,4,5	Write an algorithm of i th selection problem	4
			Question 3. a is compulsory and solve any one out of 3.b and 3.c	
Q. 3.a			Solve 0/1 Knapsack problem using Dynamic Programming (Tabulation Approach) $W=\{2,1,5,3\}$	6

			Value = {300,200,400,500} Knapsack Capacity = 5 kg Write formula. And just fill in the table. Compute maximum value obtained and items that would get selected.	
b.			Compare various approaches to find n^{th} Fibonacci series term. Which approach is better? Why?	4
c.			Write recurrence relation and time complexity of Travelling salesman problem using dynamic programming. Explain in brief why this approach has the mentioned time complexity.	4
Q. 4.a .			Create Huffman Tree and generate code for each of the given letter. Is this code prefix free? Justify. Letter - Frequency h - 1 e - 1 l - 3 o - 2 w - 1 r - 1 d - 1	6 M
b.			Kruskal algorithm is better used with sparse graphs and Prim's algorithm is helpful when dealing with dense graphs. Is this statement true? Justify.	4 M

			Question 4. a and 4 b is compulsory																									
Q. 4. a	3			6																								
b	3			4																								
			Question 5. a and 5 b is compulsory.																									
Q. 5.a	3	5, 6	Find the job sequencing/scheduling with maximum profit by considering jobs and their profits and the deadlines using greedy approach. <table><tr><td>Jobs</td><td>j1</td><td>j2</td><td>j3</td><td>j4</td><td>j5</td><td>j6</td><td>j7</td></tr><tr><td>Profit</td><td>35</td><td>30</td><td>25</td><td>20</td><td>15</td><td>12</td><td>5</td></tr><tr><td>Deadline</td><td>3</td><td>4</td><td>4</td><td>2</td><td>3</td><td>1</td><td>2</td></tr></table>	Jobs	j1	j2	j3	j4	j5	j6	j7	Profit	35	30	25	20	15	12	5	Deadline	3	4	4	2	3	1	2	6
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Profit	35	30	25	20	15	12	5																					
Deadline	3	4	4	2	3	1	2																					
b	5	1, 2	Elaborate P and NP problem classes and their relationship.	4																								
			Question 6. a and 6 b. is compulsory.																									
Q. 6. a	4	4,5,6	Prove that a problem solved using $O(n \log n)$ is polynomial time solvable or P type problem.	6																								
b	6	3,5	What is key difference between Monte Carlo and Las Vegas randomized algorithms?	4																								

CO Statements:

CO1. Formulate a given computational problem in an abstract and mathematically precise manner.

CO2. Choose a suitable paradigm to design algorithms for given computational problems.

CO3. Understand asymptotic notations and apply suitable mathematical techniques to find asymptotic time and space complexities of algorithms.

CO4. Understand the notion of NP-hardness and NP completeness and their relationship with the intractability of decision problems.

CO5. Apply randomized, approximation algorithms for given computational problems.

***Blooms Taxonomy (BT) Level No:**

1. Remembering; 2. Understanding; 3. Applying; 4. Analyzing; 5. Evaluating; 6. Creating